

Effects of hempseed and flaxseed oils on the profile of serum lipids, serum total and lipoprotein lipid concentrations and haemostatic factors.

BACKGROUND: Both hempseed oil (HO) and flaxseed oil (FO) contain high amounts of essential fatty acids (FAs); i.e. linoleic acid (LA, 18:2n-6) and alpha-linolenic acid (ALA, 18:3n-3), but almost in opposite ratios. An excessive intake of one essential FA over the other may interfere with the metabolism of the other while the metabolisms of LA and ALA compete for the same enzymes. It is not known whether there is a difference between n-3 and n-6 FA of plant origin in the effects on serum lipid profile.

AIM OF THE STUDY: To compare the effects of HO and FO on the profile of serum lipids and fasting concentrations of serum total and lipoprotein lipids, plasma glucose and insulin, and haemostatic factors in healthy humans.

METHODS: Fourteen healthy volunteers participated in the study. A randomised, double-blind crossover design was used. The volunteers consumed HO and FO (30 ml/day) for 4 weeks each. The periods were separated by a 4-week washout period.

RESULTS: The HO period resulted in higher proportions of both LA and gamma-linolenic acid in serum cholesteryl esters (CE) and triglycerides (TG) as compared with the FO period ($P < 0.001$), whereas the FO period resulted in a higher proportion of ALA in both serum CE and TG as compared with the HO period ($P < 0.001$). The proportion of arachidonic acid in CE was lower after the FO period than after the HO period ($P < 0.05$). The HO period resulted in a lower total-to-HDL cholesterol ratio compared with the FO period ($P = 0.065$). No significant differences were found between the periods in measured values of fasting serum total or lipoprotein lipids, plasma glucose, insulin or hemostatic factors.

CONCLUSIONS: The effects of HO and FO on the profile of serum lipids differed significantly, with only minor effects on concentrations of fasting serum total or lipoprotein lipids, and no significant changes in concentrations of plasma glucose or insulin or in haemostatic factors.